

Deep-Level Characterization in GaN HEMTs-Part I: Advantages and Limitations of Drain Current Transient Measurements

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Abstract—This paper critically investigates the advantages and limitations of the current-transient methods used for the study of the deep levels in GaN-based high-electron mobility transistors (HEMTs), by evaluating how the procedures adopted for measurement and data analysis can influence the results of the investigation. The article is divided in two parts within Part I. 1) We analyze how the choice of the measurement and analysis parameters (such as the voltage levels used to induce the trapping phenomena and monitor the current transients, the duration of the filling pulses, and the method used for the extrapolation of the time constants of the capture/emission processes) can influence the results of the drain current transient investigation and can provide information on the location of the trap levels responsible for current collapse. 2) We present a database of defects described in more than 60 papers on GaN technology, which can be used to extract information on the nature and origin of the trap levels responsible for current collapse in AlGaN/GaN HEMTs. Within Part II, we investigate how self-heating can modify the results of drain current transient measurements on the basis of combined experimental activity and device simulation.

Index Terms—Activation energy, deep level, gallium nitride (GaN), high electron mobility transistors, traps.

I. INTRODUCTION

GaN-based high electron mobility transistors (HEMTs) represent good candidates for next generation microwave and power electronics [1], [2]. Even though strong efforts

are being made to improve long-term reliability of HEMTs [3], [4], the dynamic performance of these devices is still affected by charge trapping phenomena [5]. One of the most evident effects of traps is the so-called current collapse, i.e., the dynamic decrease in drain current experienced by the device when operated with large gate-drain voltage swings [6]. This effect is strictly related to charge trapping at deep levels, which can be present in the as-grown material, related to the processing conditions [7], and/or induced by electrical stress.

The analysis of the properties of the deep levels located in a HEMT structure is of fundamental importance for the following aspects: on one hand, it provides information on the dependence of the dynamic behavior of the HEMTs on bias and temperature, which can be used for modeling the rf-characteristics of the devices; on the other hand, advanced techniques for defect characterization can be used for diagnostic purposes, i.e., for the identification of the nature of the deep levels generated as a consequence of electrical and thermal stress within the devices.

A quick and reliable characterization of the current collapse can be obtained through the execution of pulsed drain-current versus drain (or gate) voltage measurements (pulsed I_D - V_D or I_D - V_G characterization) [7], which allow one to investigate the changes in drain current, on-resistance, and threshold voltage induced by the capture of electrons within the AlGaN/GaN heterostructure.

Despite the effectiveness of pulsed current-voltage characterization, more accurate techniques are needed for the identification of the trap levels responsible for current collapse, and for this paper of their evolution under long-term stress. Several methods have been proposed to this aim, which are based on the analysis of a capacitance or current transient generated by the charging/de-trapping of defect levels, and can provide information on the properties (activation energy and capture cross-section) of the traps that limit the dynamic performance of the devices.

Capacitance-mode deep-level transient spectroscopy (C-DLTS) and deep-level optical spectroscopy (DLOS) have been widely employed to identify the deep levels [30]–[96]: in most of the cases, the measurements are carried out on large area Schottky diodes, since the C-DLTS can be difficultly applied to actual transistors, due to their small gate capacitance (typically lower than 1–5 pF). Unfortunately, the distribution of the electric field is significantly different in

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diodes and in actual transistors; therefore, the results obtained by analyzing diodes provide information on the properties of the deep levels, but do not give a clear indication on where those levels are located in a HEMT structure, and on how they can influence the dynamic behavior of the actual devices.

A more accurate description of the properties of the traps relevant for GaN-based transistors operation can be obtained by studying the current transients (in the ON-state) induced by exposure to a trapping voltage in the OFF-state (typically a negative gate bias, or a high drain-gate voltage), through current-mode DLTS (I-DLTS) [5], [8], [9], [10] or multiexponential transient measurements [11]–[19], [59]: the latter method is faster, since it does not require to carry out a full temperature scan, but only to measure the current transients at some (5–10) temperature levels.

Despite the relative simplicity of this technique, the choice of the parameters used for the measurements, and the method used for the analysis of the transients can strongly affect the results of this paper. More specifically: 1) several methods can be used for the extrapolation of the time constants of the charge emission processes: the multiexponential fitting [13]–[16], the analysis of the derivative of the current transients (the peak of the derivative is located in correspondence of the time constant of the emission process) [17], [18], and the stretched exponential fitting [19] have been adopted by various research groups. The choice of the method can modify the results of the investigation, since it can significantly change the shape and slope of the Arrhenius plot (see details in the following sections). 2) Current transients can be measured both in the linear [14], [16]–[18] and in the saturation region [11], [19], [20]: the choice of the operating point can significantly influence the results of the investigation, and—in some cases—hinder the observation of some trap levels (see Section III). 3) Neglecting the self-heating of the devices during the analysis of the current transients may lead to a wrong extrapolation of the Arrhenius plots. Despite the importance of these aspects, there is no systematic study in the literature describing how choice of the measurement and analysis parameters can influence the results of the drain current transient investigations. Furthermore, in many cases, the origin of the traps detected by current transient measurements is not univocally identified, due to the difficulty of comparing the signatures of the deep levels with those described in previous literature reports.

The aim of this paper is to contribute to the understanding of the properties of the traps that limit the performance of GaN-based HEMTs. 1) On one hand, we demonstrate that the choice of the measurement and analysis parameters (such as the voltage levels used to monitor the current transients, the duration of the filling pulses, and the method used for the extrapolation of the time constants of the capture/emission processes) can significantly influence the final results of the drain current transient investigations. 2) Furthermore, we demonstrate that through the adoption of proper trap filling voltages, it is possible to use isothermal drain-current transient investigation to identify the location of the traps responsible for current collapse. 3) We demonstrate that by comparing the trap

signatures extrapolated by drain-current transient investigation (i.e., activation energies and capture cross-sections) with literature data it is possible to formulate hypotheses on the origin of the traps responsible for current collapse. To this aim, we present a database of the trap levels detected in GaN and AlGaIn diodes, and AlGaIn/GaN HEMTs, which was constructed starting from the data reported in more than 60 scientific papers. This database will constitute a useful reference for researchers involved in the investigation of the trapping processes in GaN transistors, since it allows to directly correlate the signatures of trap states extrapolated by experimental investigation with the data presented in a wide number of literature reports.

II. EXPERIMENTAL DETAILS

This paper was carried out on HEMTs grown by MOCVD on a SiC substrate. Device structure consists of a 1.8- μm GaN buffer layer, with a 22-nm-thick AlGaIn (18% Al) barrier. Gate-source spacing, gate-drain spacing, gate length, and gate width are, respectively, 1, 3.5, 0.5, and 100 μm . The devices have a 1.3- μm source-connected field-plate. To quantify current collapse, we carried out pulsed I_D – V_D and I_D – V_G characterization, starting from several quiescent bias points, as described in [7]. The analysis of the traps responsible for current collapse was carried out by drain current transient investigation. For these measurements, the drain was connected to a power supply (V_{DD}) through a load resistor: the device was kept in a defined trapping condition (V_{GF} ; V_{DF}) for 100 s (in the OFF-state), and the subsequent transient investigation was carried out by switching the gate terminal (from the OFF-state to the ON-state), and by recording drain current response over eight time decades (from 1 μs to 200 s) by means of an oscilloscope. Further details on the parameters used for the investigation are given in the following sections.

III. CURRENT COLLAPSE AND TRANSIENT INVESTIGATION

A. Pulsed Measurements

The presence of trapping can be quickly analyzed by pulsed current-voltage characterization, starting from several quiescent bias points (V_{GQ} , V_{DQ}). In a previous research work [7], we demonstrated that traps located in the region under the gate or in the gate-drain access region can be separately identified by pulsed I_D – V_D and I_D – V_G characterization. Samples with large density of defects located below the gate region, exhibit predominantly threshold voltage (V_{TH}) dynamical shift, since the net polarization charge and related 2DEG sheet charge density is affected by negative charge trapped under the gate [21]. On the other hand, the presence of traps in the gate-drain access region determines a dynamic decrease in the transconductance peak, with no significant modification of V_{TH} (Fig. 2).

The devices analyzed within this paper have been characterized by means of the method described in [7]: as shown in Fig. 1, for the (V_{GQ} , V_{DQ}) = (–6 V; 50 V) quiescent bias point, we measured a 40% decrease in I_{DSS} [Fig. 1(a)],

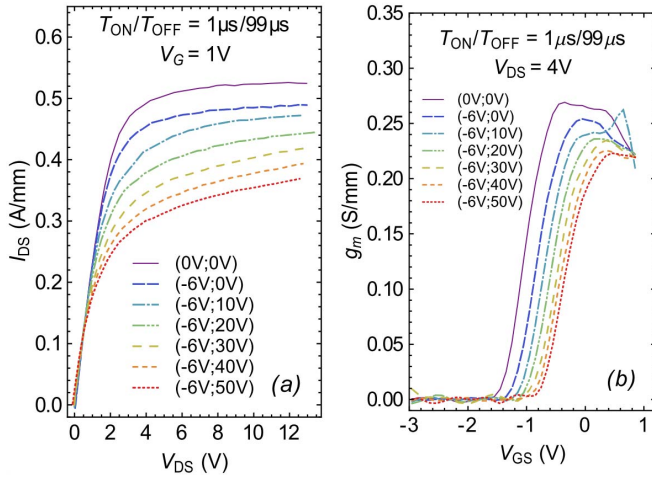


Fig. 1. (a) I_D - V_G and (b) g_m (derived from I_D - V_G) pulsed characterization: the dynamic current collapse is mainly caused by V_{TH} positive shift suggesting charge trapping within the epilayers in the intrinsic region below the gate contact. Employed pulse width/period ratio is 1 μ s/100 μ s.

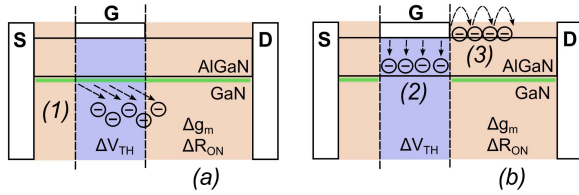


Fig. 2. Possible trapping mechanisms and their influence on electrical parameters in (a) semi-ON-state and (b) OFF-state bias operating points.

associated with a remarkable V_{TH} positive shift [700 mV, Fig. 1(b)], slight R_{ON} increase, and 20% g_m peak reduction. Results indicate that, since the current collapse is mainly related to V_{TH} positive shift, the deep levels which cause the majority of dynamic performance dispersion—for this case study—are located under the gate region, into the semiconductor epilayers.

B. Drain Current Transients: Choice of the Bias Points in the ON- and OFF-State

Thanks to the fact that it explores the whole output and transfer characteristics, pulsed I_D - V_D and I_D - V_G characterization is very effective for the investigation of current collapse; however, since its time acquisition window is fixed at 1 μ s, it does not provide information on the dynamic behavior of the trap states responsible for the observed current collapse, and the subsequent identification of the corresponding deep-levels. For this reason, an extensive drain current transient analysis, which can explore the capture and emission kinetics from the micro-second to the hundreds of second time scale, has been performed by following the methodology described in Section II.

The measurements can be carried out by analyzing the charge de-trapping transients either in the linear or in the saturation region. Experimental data collected within this paper (Fig. 3) indicate that the choice of the bias point used for the measurement of the de-trapping transient can have a significant

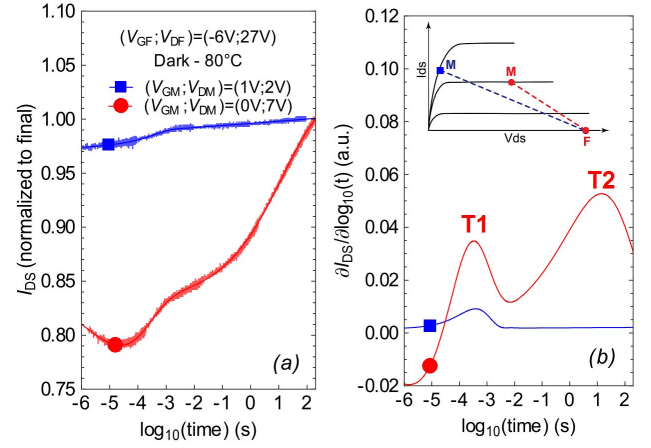


Fig. 3. (a) Drain current transients performed in the linear and saturation regions and (b) related differential signals: in good agreement with pulsed measurement, transients recorded in linear region detect only a weak emission process, while those recorded in saturation region reveal much higher current collapse and the presence of two emission processes-labeled T1 and T2.

impact on the results: for the analyzed devices, the transients recorded in linear region detect only a weak emission process (blue curve in Fig. 3), while those recorded in saturation region (red curve in Fig. 3) reveal a remarkable current collapse and the presence of two emission processes, labeled, respectively, “T1” and “T2.” This result is very well correlated with that obtained with pulsed measurements (Fig. 1), which indicate that current collapse is mainly caused by threshold voltage shift, hence manifesting itself predominantly in saturation region.

In the following, we demonstrate that through the use of proper trap filling voltages, it is possible to achieve information on the location of the traps. This result was obtained by carrying out a set of transient measurements, starting from three different filling bias conditions: 1) (V_{GF} ; V_{DF}) = (-6 V; 27 V), in the OFF-state, with both high negative V_{GS} and high positive V_{DS} , which promotes both gate- and drain-dependent trapping; 2) (V_{GF} ; V_{DF}) = (-1 V; 27 V), in the semi-on state, with high V_{DS} level, which promotes only the drain-dependent trapping mechanisms; and 3) (V_{GF} ; V_{DF}) = (-6 V; 8 V), in the OFF-state, with high negative V_{GS} and low V_{DS} to probe predominantly the gate-dependent trapping mechanisms.

The results of this comparison are summarized in Fig. 4, and indicate the presence of several de-trapping transients: the amplitude of the transient labeled as “T1” is enhanced by filling pulses with high drain voltage, and experiences its maximum amplitude under semi-on trapping conditions (V_{GF} ; V_{DF}) = (-1 V; 27 V). On the contrary, the broad spectrum of transient “T2” is constituted by two components: one drain-influenced (labeled T2D), which is enhanced during semi-on condition, and one gate-influenced (labeled T2G), which is enhanced during off-state trapping with moderate drain voltages and high negative gate bias (V_{GF} ; V_{DF}) = (-6 V, 8 V). Furthermore, during semi-on trapping condition, an additional weak electron-capture or hole-emission process (labeled T3) is detected.

The results described in Fig. 4 therefore indicate that by choosing proper filling voltages, it is possible to distinguish

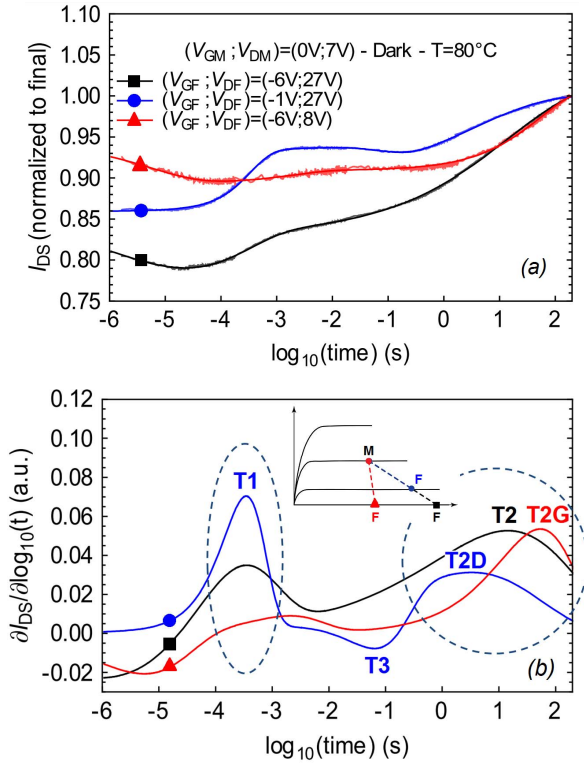


Fig. 4. (a) Drain current transients recorded after different trapping conditions and (b) related differential signals: T1 amplitude is enhanced by the combination of drain voltage and drain current experiencing his maximum after semi-on trapping condition, whereas broad T2 spectrum is composed by one drain-influenced component (T2D) and one gate-influenced component (T2G).

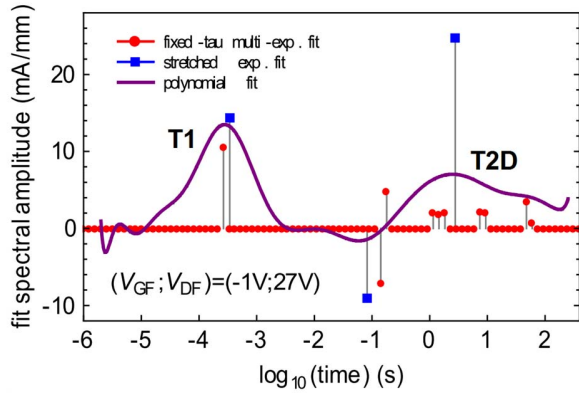


Fig. 5. Comparison among the three data fitting solutions: though the pure-exponential T1 process is correctly detected by all the fitting methods, the nonexponential T2D signal is properly detected only by stretched exponential fit, since its detection by polynomial fit and fixed-tau-multiexponential fit is affected by spectral dispersion.

between drain- and gate-dependent trapping processes: the peaks T1 and T2D—whose amplitude is maximized by the combination of high drain voltage and relatively high drain current levels—can be ascribed to the trapping of negative charge, likely promoted by hot-electrons, in the buffer region [see case (1) in Fig. 2(a)]. On the other hand, trap T2G is supposed to be located in the region under the gate, where charge trapping could be promoted by high reverse gate

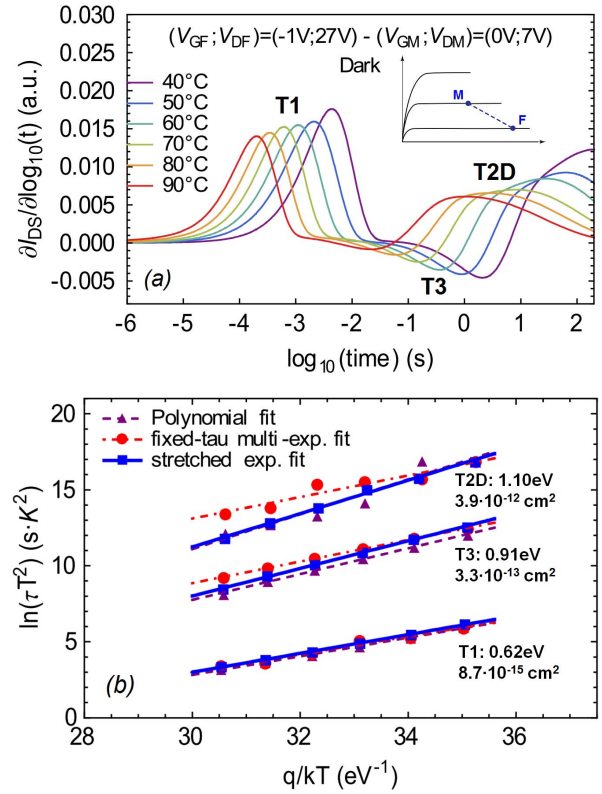


Fig. 6. (a) Thermal activation and (b) Arrhenius plot with apparent activation energies (E_A) and capture cross-sections (σ_c) extrapolated by stretched exponential fit of T1-T2D- and T3-related de-trapping processes. Comparison with other fitting methods is also reported.

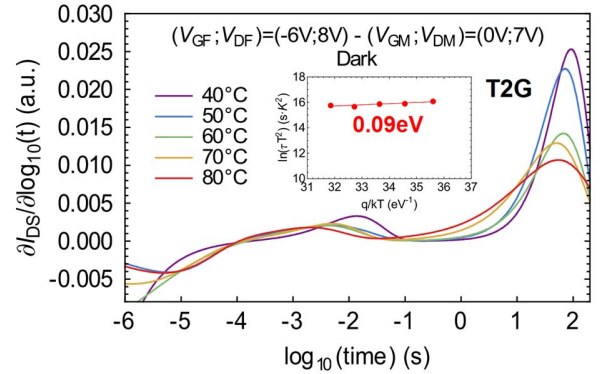


Fig. 7. Thermal activation of T2G signal: very low activation energy and detrapping time constant in the order of 100 s have been identified.

voltage and the subsequent high gate leakage current levels [see case (2) in Fig. 2(b)]. The superficial trapping phenomena [case (2)], reported for completeness in Fig. 2(b), are unlikely to occur in these devices since only slight extrinsic parameters dynamical degradation is detected. These inferences will be confirmed in the following sections, where further hypotheses on the nature of these traps are presented. Furthermore, since the T2-related processes have typical time constants close to the upper boundary of the acquisition window, further assessment measurements (not shown here) have been performed up to 1 ks to confirm the deep-levels features described throughout the manuscript.

C. Methods for the Extrapolation of the Time Constants of the Current Transients

The calculation of the time constants of the current transients is a necessary step toward the extrapolation of the Arrhenius plot of the trap levels. Within this paper, we implemented and compared three different methods for the evaluation of time constants: 1) the fitting of transient data by a polynomial function, and the subsequent extrapolation of time constants from the peaks of the $\partial I_{DS}/\partial \log(t)$ (derivative) curves [17]; 2) the fitting of the data by a function represented by the sum of 100 exponentials with fixed time constants and variable amplitude coefficients [14]; and 3) the fitting of the data by the stretched multiexponential function

$$I_{DS}(t) = I_{DS,final} - \sum_i^N A_i e^{-\left(\frac{t}{\tau_i}\right)^{\beta_i}} \quad (1)$$

where the fitting parameters A_i , τ_i and β_i are, respectively, the amplitude, the typical time constant and the nonexponential stretching-factor of the N detected charge emission ($A_i > 0$) or capture ($A_i < 0$) processes (N typically ranges between 2 and 4 depending on the analyzed samples). The comparative results of the different fitting algorithms are summarized in Fig. 4. As can be noticed, the three methods provide different behaviors: in the ideal case of almost purely exponential transient (trap T1 in Fig. 5), the three methods return similar fitting parameters (amplitudes and typical time constants), hence leading to the same Arrhenius plot [see the Arrhenius plot of trap T1 in Fig. 6(b)]; conversely, in the presence of stretched nonexponential processes (e.g., T2D emission in Fig. 5) the output of both polynomial and fixed-tau multiexponentials fitting methods are affected by spectral dispersion, which manifest itself as the distortion of the differential current peak (for the polynomial fit case) or the splitting of spectral lines (for the multiexp. fit). As a consequence, the time constants extrapolated by means of polynomial fitting and fixed-tau multiexponential can significantly differ from those obtained through stretched exponential fitting; this difference may lead to a wrong extrapolation of the Arrhenius plots [see the Arrhenius plot of trap T2D in Fig. 6(b)]. Based on these considerations, the investigation described in the following was carried out by means of stretched exponential fitting, since the slow T2-related signals are effectively characterized by nonexponential behavior.

Previous literature reports ascribed the presence of strongly nonexponential transients to the following causes: 1) current transients originate from the superposition of different deep-levels emission processes [22]: as shown in Fig. 4, T2 is indeed composed by the namely T2D (drain-dependent) and T2G (gate-dependent) trap states (Fig. 4); 2) current transients are related to the presence of high density of defect states, as in the case of extended line defects, that lead to trapped electron delocalization and formation of mini-bands [23], [24]; 3) current transients originate from the presence of defect states in alloy materials, such as $\text{Al}_x\text{Ga}_{(1-x)}\text{N}$, which lead to deep-level splitting due to band discontinuity [25]; 4) the charge emission parameters smear due to the nonuniform distribution of the electric field within the complex HEMT

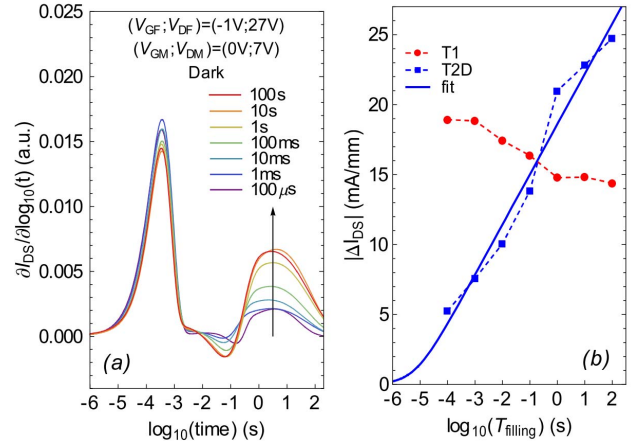


Fig. 8. (a) T1 and T2D typical spectrum and (b) related process amplitudes of drain current transient recorded after different filling times, supporting the hypothesis that T1 originates from point defects, and T2D is related to extended line defects, which lead to capture barrier formation.

structure [23]; and 5) trapping/de-trapping kinetics are not governed by thermal emission, but by other mechanisms such as hopping or trap-assisted-tunneling [19], [26].

IV. IDENTIFICATION OF THE DEEP LEVELS

A. Trap Signatures

For a better understanding of the properties of the deep levels responsible for current collapse, it is necessary to extrapolate the Arrhenius plots of the traps and their signature in terms of activation energies and capture cross-sections. These data are obtained by carrying out the current transient investigation at several temperature levels. The results of this investigation are summarized in Figs. 6 and 7: the signatures of T1, T2D, and T3 (Fig. 6) correspond to apparent activation energies and apparent capture-cross sections of 0.62 eV/ $8.7 \times 10^{-15} \text{ cm}^2$ for T1; 1.10 eV/ $3.9 \times 10^{-12} \text{ cm}^2$ for T2D; and 0.91 eV/ $3.3 \times 10^{-13} \text{ cm}^2$ for T3. Conversely, T2G experiences very low thermal activation ($< 0.1 \text{ eV}$) and typical time constants in the order of 100 s (Fig. 7). In order to avoid results misinterpretation, the actual device temperature has been estimated taking into account the dissipated power during the measurement phase (around 1.5 W/mm) and the thermal resistance of the device (10 K mm/W) measured by means of the pulsed-current method described in [27]. Further details on the importance of temperature correction are extensively described in Part II.

B. Filling Time Dependence

In the following, we demonstrate that it is possible to obtain more detailed information on the characteristics of the traps described above, by investigating the dependence of the signal related to traps T1, T2D, and T3 on the duration of the filling pulse used to promote the charge trapping.

For this investigation, trapping condition [$(V_{GF}; V_{DF}) = (-1 \text{ V}; 27 \text{ V})$] and temperature (80 °C) were kept constant, whereas filling time has been varied from 100 s down to 100 μs. As described in Fig. 8, the amplitude of T2D displays

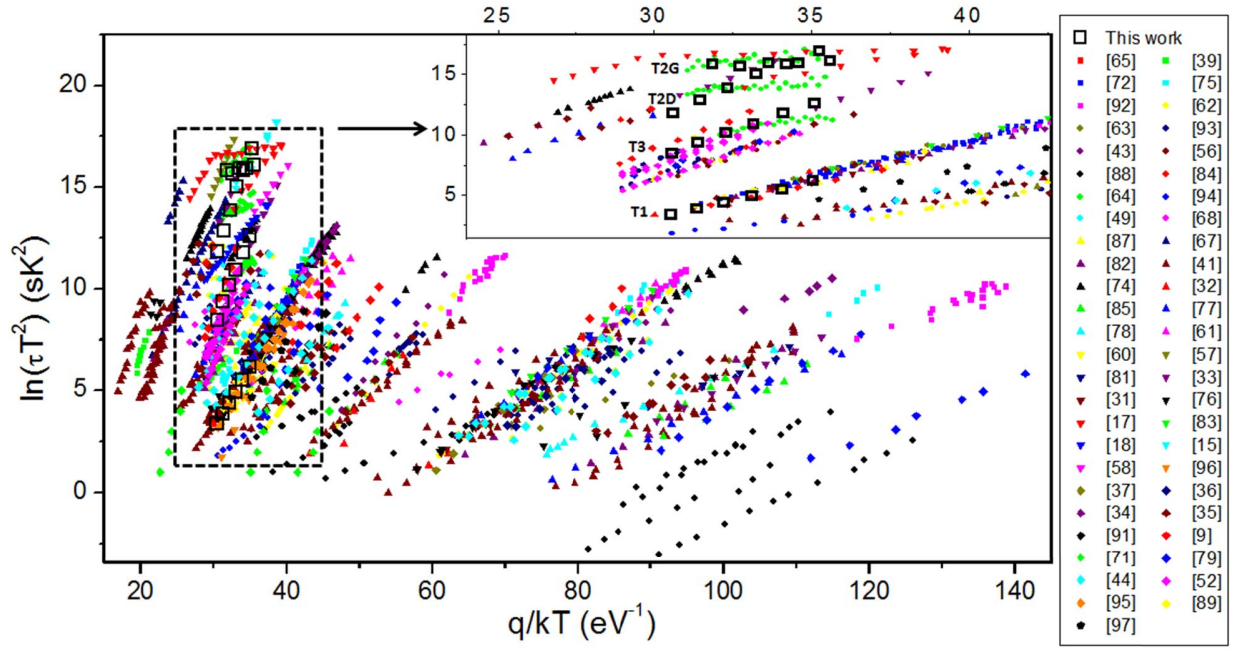


Fig. 9. Arrhenius plot of the deep levels in GaN- and AlGaIn-based layers and devices, collected in the database realized in this paper.

a logarithmic dependence on the duration of the filling pulse, suggesting that the corresponding trap level may be tentatively related to the presence of highly localized defect states, such as dislocations. In fact, as described in [28]–[30], the charging of either dislocation core sites or hypothetical point defects and impurities segregated around the dislocations, would cause a constant reduction of the capture rate due to the formation of a Coulomb capture barrier with the increase of negative trapped charge. According to [30], the dependence of the charge filling time (before the saturation of the filled trap states) can be expressed as

$$n_T(t_F) = c_n \tau N_T \ln \left(1 + \frac{t_F}{\tau} \right) \quad (2)$$

where c_n , N_T and τ represent respectively the capture probability, the defect states density and the characteristic time for the capture barrier build-up. On the other hand, the amplitude of peak T1 exhibits opposite behavior, hence suggesting that the corresponding trap level is related to point defects; the slight decrease of its amplitude with the increasing of filling time may be due to the partial charge emission from T1-related trap, since the filling time becomes larger than its typical emission time constant (in the milliseconds range) while the overall trapping force is lowered by the rising of T2D signal.

C. Identification of Defects by a Database of Deep Levels

The understanding of the physical origin of a trap is not straightforward, even if the activation energy and cross section of the corresponding deep level have been accurately extrapolated. To compare the results obtained within this paper with previous literature reports, we have constructed a database of the deep levels presented over the last decades in more than 60 journal papers. Fig. 9 and Table I report a summary of the

catalogued deep levels: for each range of activation energies we list the reference papers, the type of analyzed samples, and one or more hypothesis on the origin of these traps as reported by the authors.

To demonstrate that the presented database can be used to achieve a more detailed description of the properties of the deep levels, in Fig. 9 we also report the comparison between the signatures of the deep levels detected within this paper and previous literature references. As can be noticed, the signature of trap T1 ($E_A = 0.62$ eV; $\sigma_c = 8.7 \times 10^{-15}$ cm²) reveals good matching with the level $E_2/E_C = 0.60$ eV, which is broadly detected in several works regarding both GaN Schottky diode and HEMT structures [14], [30]–[45], [73]. This level has been generally ascribed to GaN point defects, even though its origin is still controversial: Nitrogen anti-site native point defect [31], [34] and foreign impurities and dopants (C [40], Si [47], Mg-H complex [32], Fe [96]) have been also reported. Level T2D ($E_A = 1.10$ eV; $\sigma_c = 3.9 \times 10^{-12}$ cm²) reveals similar signature of deep levels associated with GaN dislocations [9], [33], [46], [49], [50]. This would be consistent with the logarithmic filling time dependence (Fig. 8), which—as described above—suggest that T2D may be related to extended defects. Level T3 ($E_A = 0.91$ eV; $\sigma_c = 3.3 \times 10^{-13}$ cm²) displays similar signature with V_{GA} -related deep acceptor states reported in [38], [48], [51], and [52] (in this case the 0.91 eV apparent activation energy represent the trap level energy from the valence band maximum). Nonetheless, $E_V + 0.9$ eV was also reported more recently in [53] as possible charge-state transition of Carbon impurities in Nitrogen substitutional position (C_N). Finally, the very low thermal activation and the very slow de-trapping typical time constant of the gate-influenced T2G are similar to the trap signature detected in HEMT structure and reported in [17] and could be supported by these

TABLE I
DATABASE OF THE DEEP LEVELS IN GaN-AND AlGaN-BASED LAYERS AND DEVICES

Reference papers	Analyzed samples	Deep level energy (eV)	Interpretations
Umana-Membreno [95], Soh [30], Park [83], Cho [36], Johnstone [89], Cho [37], Choi [62], Arehart [43], Cho [86], Chen [45] Chen [45] Lee [77] Polyakov [91] Gassoumi [72], Okino [39] Heitz [71] Umana-Membreno [95] Soh [30] Caesar [18], Tapajna [17], Lee [77] Cho [37], Cho [36], Umana-Membreno [75], Stuchlikova [46], Ashraf [61], Arehart [43], Chung [34], Chen [45], Fang [35] Polyakov [51] Cardwell [97] Chen [47] Hacke [32], Hierro [66] Stuchlikova [46] Jhonstone [89] Okino [39] Silvestri [96] Asgar [68] Polyakov [51], Calleja [57] Fang [35] Asgar [68] Auret [33], Fang [49], Fang [50] Stuchlikova [46] Arehart [43], Arehart [44], Arehart [9], Hierro [66] Sasikumar [5] Zhang [81] Aggerstam [74] Arehart [43], Arehart [44], Hierro [66] Arehart [56] Sasikumar [94], Arehart [43], Hierro [66] Zhang [81], Arehart [43], Sasikumar [5], Arehart [44], Henry [93] Sasikumar [5], Arehart [9] Arehart [56]	Various GaN-based devices n-GaN TMGa GaN p-GaN AlGaIn/GaN HEMT Fe doped GaN n-GaN Si doped GaN Various GaN-based devices Various GaN-based devices Fe doped GaN AlGaIn/GaN HEMT n+p GaN diode Mg doped p-type GaN AlGaIn/GaN HFET n-type GaN AlGaIn/GaN MIS-HEMT Fe-doped AlGaIn/GaN HEMT GaN pn diode Fe-doped or U.I.D. GaN n-type GaN on sapphire GaN pn diode Various GaN-based devices AlGaIn/GaN HFET Various GaN-based devices AlGaIn/GaN HEMT on SiC c/m plane GaN Fe doped GaN Various GaN-based devices AlGaIn (Al 30%) Various GaN-based devices Various GaN-based devices AlGaIn/GaN HEMT AlGaIn (Al 30%)	EC - 0.09/0.27 EC - 0.12 EC - 0.14 EC - 0.15 EC - 0.3/0.34 EC - 0.34 EC - 0.355 EC - 0.37/0.4 EC - 0.44/0.49 EC - 0.5/0.62 EC - 0.5 EC - 0.57 EC - 0.59 EC - 0.6/0.62 EC - 0.6/0.64 EC - 0.613 EC - 0.68 EC - 0.72 EC - 0.76 EV + 0.85, 0.94 EC - 0.89 EC - 0.96 EC - 0.95/1.02 EC - 1.118 EC - 1.28/1.35 EC - 2.3 EC - 2.47/2.49 EV + 2.5/3 EC - 2.6/2.64 EC - 3.11 EC - 3.2/3.22 EC - 3.24/3.31 EC - 3.7/3.76 EC - 3.93	Nitrogen vacancies Surface Carbon or hydrogen impurities Mg ionization Possibly AlGaIn surface Fe3+/2+ Mg impurities Si dopant C/O/H impurities, possibly in nitrogen substitutional position Nitrogen antisites Fe dopant influenced by Fe dopant Si dopant Mg-H complex formation VGa + Oxygen complex Nitrogen vacancies Surface Fe dopant Nitrogen interstitials Gallium vacancy Nitrogen interstitials Gallium vacancy or N interstitials Threading dislocations VGa + Oxygen complex Carbon interstitial defect Surface VGa + Hydrogen complex Fe dopant VGa or VGa-H or VGa-2H Cation vacancy Residual Mg acceptor CN substitutional Mg or C substitutional in AlGaIn Mg impurities

explanations: 1) electron transport mechanism dominated by **RC discharging processes** [13], [14]; 2) **superficial shallow states filled and emptied by hopping conduction mechanisms** [26], [55], [67]; or 3) **deep-levels charge de-trapping by electron trap-assisted-tunneling mechanisms** [14], [15], [54], [60], [70].

D. Hypothesis on Trap Location

For what concerns the location of traps, both T1 and T2D are supposed to be located in the GaN buffer, for the following reasons: 1) **both T1 and T2D filling trap mechanisms are promoted at high drain voltage levels or during semi-on state, suggesting hot-electrons-assisted filling mechanism** (Fig. 4) and 2) trap signatures match with GaN-related point defects (T1) and dislocations (T2D). On the contrary, results regarding T2G may suggest the presence of defect states located in **AlGaIn barrier [56], where trap filling is promoted by high negative bias probably due to gate leakage current (Fig. 4), and where the subsequent electron emission toward the 2DEG or the gate contact may be dominated by hopping or trap-assisted tunneling mechanisms.**

Hypothesis on the localization of trap states are supported by pulsed I_D-V_D and I_D-V_G measurements, where the majority of the current collapse is related to V_{TH} positive shift. The slight R_{ON} increase and g_m reduction may be explained by the presence of the defects not only under the gate, but also in access regions where can be filled, in the proximity of gate edges, by the same trapping mechanism described above.

V. CONCLUSION

Within this paper, we have presented an extensive investigation of the advantages and limitations of the pulsed and transient methods used for this paper of the deep levels responsible for current collapse in AlGaIn/GaN HEMTs. We have demonstrated that the choice of measurement and analysis parameters can strongly influence the results of the investigations, and we have provided guidelines for the execution of accurate and correct measurements. More specifically, we have investigated how the choice of the measurement bias point, the voltage and duration of the trap filling pulses, and the method used for the extrapolation of the time constants can impact the results of the investigations. Furthermore, we have presented a database of the deep levels in GaN-based devices: based

on the experimental data obtained within this paper, we have shown that this database can be used for the identification of the physical origin of the trap states responsible for current collapse in AlGaIn/GaN HEMTs.

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